

1. Executive Summary

This report looks at over 12,000 customer service interactions to answer a key question: can we tell in advance when a customer complaint is likely to escalate into something serious? The data covered four possible outcomes, no escalation (48.9%), minor escalation (30.4%), serious escalation (16.7%), and legal escalation (4.0%). Several predictive models were tested, and the best performer, Gradient Boosting, correctly identified the likely outcome 96% of the time. To understand why, a tool called SHAP was used to highlight which factors mattered most. Across the most serious cases, the clearest warning signs were: an existing risk indicator on the customer's account, service pressure score, customer engagement score, wait time, total burden, and sentiment score, with the relative importance varying depending on whether a serious or legal outcome was being predicted. When the data was grouped without prior assumptions, one natural cluster stood out, with 55% going on to have a serious escalation and 22% reaching a legal dispute. The key takeaway is that escalation is predictable and largely preventable. Customers who escalate tend to have a history of poor service experiences building up over time, meaning there is a real opportunity to step in early before situations reach a serious or legal stage.

2. Introduction

The goal of this analysis is to provide insight into whether customer escalations can be predicted before they happen, and to identify what factors drive them. This report applies supervised machine learning and unsupervised clustering to build an early warning system for escalation risk. The report walks through data exploration and visualisation, pre-processing and feature engineering, model selection and evaluation, and clustering analysis. This report is structured as follows: Advanced Data Analysis Report, Jupyter Notebook (Appendix A), Data Dictionary (Appendix B), and Travel Log (Appendix C).

3. Data Exploration, Visualisation, and Dictionary

3.1 Data Exploration

The raw dataset contains 12,362 rows and 41 columns, representing individual customer interactions. The target variable is the escalation level and has four classes including: No escalation (48.9%), minor escalation (30.4%), serious escalation (16.7%), and legal escalation (4.0%). A random sample of 10 records was inspected to get a sense of the data, confirming the presence of both categorical and numerical attributes ranging from customer demographics to operational performance metrics. The dataset initially comprises 15 object (categorical) columns, 7 integer columns, and 19 float columns. Several columns have missing values: sentiment score (5.10%), customer engagement score (6.68%), agent experience years (1.80%), subscription value str (3.23%), and risk indicator and service pressure score each have around 1.46-1.51% missing. The highest age value is 663, which is an obvious outlier error. The wait time seconds

max is 1520s (approx. 25 minutes), and resolution time minutes max is 470. These may be potential outliers. The region column has 47 unique values, suggesting there may be errors due to high cardinality or noise.

Based on the info retrieved and the column structure, the dataset's 41 features can be grouped as follows: **Target Variable:** escalation level (object): No escalation, Minor escalation, Serious escalation, and Legal escalation. **Customer Demographics:** age (float64): customer age in years; tenure months (int64): number of months the customer has been with the service; loyalty status (object): membership tier (Bronze, Silver, Gold, Platinum); segment (object): customer segment (e.g., Retail, Student, Enterprise). **Interaction Context:** region (object): geographic region of the customer; industry (object): industry sector; channel (object): contact channel (Phone, Chat, Email, In-app); issue category (object): broad category of the issue (e.g., Billing, Technical, Account, Orders, Returns); issue subtype (object): specific sub-category of the issue; product line (object): product type; device type (object): device used; time of day bucket (object): time of day when contact was made. **Operational Metrics:** wait time seconds (int64): time spent waiting before agent connection; resolution time minutes (int64): time taken to resolve the issue; previous interactions 30d (int64): number of contacts in the past 30 days; agent experience years (float64): years of experience of the handling agent; first contact resolution (object): whether the issue was resolved on first contact (Yes/No); number of transfers (int64): number of times the call or chat was transferred; SLA breaches last 6m (int64): number of SLA breaches in the past six months; lifetime tickets (int64): total historical tickets raised by the customer; average response delay (float64): average delay in agent responses. **Customer Value and Risk:** sentiment score (float64): sentiment of the interaction; monthly fee (float64): customer's monthly subscription fee; subscription value str (object): string representation of subscription value; churn risk band (object): categorical churn risk (e.g., Low, Medium, High); priority flag (object): whether the customer is flagged as high priority (Yes/No). **Behavioural and Telemetry Features:** behaviour score 1 through behaviour score 5 (float64): five composite behavioural indicators scored 0-100; ops telemetry kpi 1 through ops telemetry kpi 5 (float64): five standardised operational KPI metrics. **Derived Scores:** post interaction quality score (float64): quality score assigned after interaction; risk indicator (float64): a computed escalation risk signal; service pressure score (float64): a measure of service load or pressure; customer engagement score (float64): a composite engagement metric.

Initial inferences at this stage include customer engagement score being identified as the most powerful individual predictor, and operational service metrics such as wait time, number of transfers, monthly fee, agent experience, and SLA breaches last 6m being weaker individual indicators. This is notable as it challenges that these poor service operations directly drive escalation. However, these should not be ruled out completely. The exploration reveals an imbalanced dataset, and targeted data quality work on missing values and extreme outliers is needed before modelling begins.

3.2 Visualisations

The escalation level distribution (**Appendix A, Figure 1**) count plot confirms the class imbalance visually. The bar for “No escalation” is notably larger than “Minor escalation”, which overshadows “Serious escalation” and “Legal escalation”. The missingness heatmap (**Appendix A, Figure 2**) shows missingness correlation across columns, which variables tend to be missing together. The critical finding is a 0.9 missingness correlation between sentiment score and customer engagement score. A correlation this high means these two fields are almost always missing together for the same set of records. The loyalty status also shows a slight correlation (0.1) with age missingness. These two columns are missing largely independently of each other. The boxplots showing some numeric features by escalation (**Appendix A, Figure 3**) compare wait time seconds, number of transfers, monthly fee, agent experience years, and SLA breaches last 6m across escalation levels. The inference here is that these metrics are not strong for individual predictors of escalation severity because it shows nearly identical distributions across escalation levels.

Next are the boxplots showing sentiment and engagement by escalation level (**Appendix A, Figure 4**). For sentiment score, a flat median is displayed across all classes (around 0). Surprisingly, customer sentiment seems not to cleanly separate escalation levels. This may reflect that sentiment is captured after the issue is partially resolved. For customer engagement score, there is a clear increasing trend with escalation severity. Customers with No escalation have a median around 4, Minor escalation around 7, Serious escalation around 10, and Legal escalation around 13. This is a clear numeric signal of escalation risk found in the exploration and suggests this feature will be highly predictive. In other words, for customer engagement score, the feature distribution for each class is distinct from one another. For sentiment score, the classes do not separate cleanly. For example, if a customer’s sentiment score is -0.3, you cannot deduce which escalation class they belong to because all four classes have their median at roughly 0. The distributions are overlapping almost completely. Whereas, for engagement, a score of 13 means it is common in legal escalation cases and rare in no escalation cases.

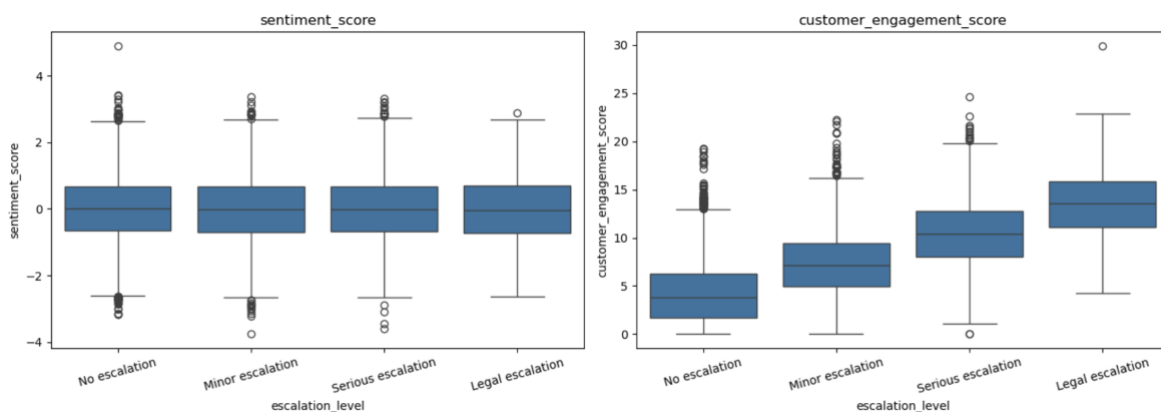


FIGURE 4: SENTIMENT AND ENGAGEMENT VS ESCALATION LEVEL

The histogram displaying behaviour scores (**Appendix A, Figure 5**) shows approximately a normal bell-shaped distribution centered around 40-55 with a range of 0-100. They appear to have a similar variance (approx. 15 std). Since this seems to appear identically distributed, a check to see whether they are independent or correlated before using as features would be appropriate.

3.3 Data Dictionary

A data dictionary table was created to summarize each column's data type, not-null count, null count, null percentage, and unique value count. This table serves as a structured reference throughout the analysis. Other attributes are appended after data cleaning and transformation to show new and accurate data.

4. Data Cleaning, Feature Engineering and Pre-Processing

4.1 Data Cleaning and Pre-Processing

For data types, subscription value str stores monetary values as strings, which cannot be used in modelling. To fix it, `.str.replace()` strips the currency symbol and commas using regex, then `.astype(float)` converts the cleaned string to a number. The column is then renamed to subscription value. Next, the first contact resolution column is stored as a generic object type. It is converted to a binary integer by mapping "Yes" to 1 and "No" to 0, with the resulting counts showing 8,922 positive resolutions and 3,440 negative ones. This will allow calculating a resolution rate with mean, use it in a model, or aggregate it easily. Finally, the priority flag column is verified whether it is binary (Yes/No), revealing 10,067 "No" entries and 2,295 "Yes" entries, confirming that it could similarly be encoded as a binary variable for further analysis.

Two columns need string cleaning: region and issue subtype. Both are object columns with inconsistent spellings, typos, and near-duplicates. The "thefuzz" library fuzzy matches each value against a list of valid entries and replaces it with the closest match. Regions (London, South East, Wales, Scotland, North West, Midlands) are stored as valid regions; and issue subtypes are stored in valid issue subtypes with the cleaned entries. After correcting the two columns, a check for duplicates is applied, and duplicates are dropped to eliminate redundancy.

Numeric columns were checked for outliers using a function with the IQR method. Rather than removing rows, extreme values were capped at the IQR bounds ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$) to preserve records while limiting the influence of extremes on model training. This was applied to the majority of numeric columns, including operational metrics such as wait time seconds, number of transfers, and SLA breaches last 6m. Three columns were left uncapped including tenure months, agent experience years, and age (after the >120 correction), as outliers in these represent plausible values.

Several columns had missing values. For categorical columns (loyalty status, time of day bucket), for which mean and median are not applicable, had missingness below 0.2% and were

imputed using the mode, the most frequently occurring value. Numeric columns (age, agent experience years, risk indicator, service pressure score, subscription value) had missingness between 0.25% and 1.8% and were imputed using the median, which is less sensitive to extreme values than the mean. Sentiment score and customer engagement score required separate handling. A missingness correlation of 0.9 indicates they go missing together, likely from the same data source. A binary flag was created before imputation to preserve this signal. A record with no sentiment data may reflect an abruptly ended interaction or system failure, both of which could correlate with escalation outcome. It is confirmed that 11,422 rows had complete data for both sentiment score and customer engagement score; 818 rows had at least one of those values missing and were flagged with a 1. Those 818 records are now imputed, but the flag preserves the fact that their sentiment data was originally absent.

To address leakage risks, post interaction quality score, resolution time minutes, and first contact resolution were dropped, as all three are known once the interaction has concluded and would not be available at prediction time. Checking the target variable distribution confirmed class imbalance. Once more, nearly half of the dataset (48.9%) belongs to the no escalation class, while legal escalation accounts for only 4% of the data.

4.2. Feature Engineering

Three features were engineered from existing variables. The total burden variable adds number of transfers and SLA breaches into a single column. Both measure operational failure. For example, a customer being passed around and deadlines being missed. Combining them produces one score representing how badly an interaction was handled. The high-risk variable is a binary flag set to 1 only when a customer is flagged as high churn risk and their interaction is marked as priority, identifying customers who are already at risk of leaving and have been internally flagged as important. The recent interaction ratio variable divides contacts in the last 30 days by lifetime tickets, capturing whether a customer's contact frequency has increased relative to their history. For example, a customer who has made 10 contacts in the last 30 days out of 11 lifetime total is going to behave differently than one with 10 contacts out of 500.

Since models require numeric input, categorical columns were encoded before training. The target variable, escalation level, was encoded into integers 0 through 3 using LabelEncoder to convert the four classes into numeric form. Priority flag was already confirmed as binary integer during the data type fixing step. The remaining nominal columns including channel, issue category, issue subtype, product line, device type, region, industry, segment, time of day bucket, churn risk band, and loyalty status were encoded using `pd.get_dummies`, which creates a separate binary column for each category. Redundancy is avoided by removing one category per column. When all remaining dummy columns are 0, the models infer to the dropped category by elimination.

4.3. Train/Test Split and Scale (Oh My!)

This is where the train/test split happens on the cleaned dataset. The total cleaned dataset has 12,240 rows since duplicates and problematic rows were removed during cleaning. After importing the necessary sklearn libraries, X is defined as all feature columns and y as the encoded escalation level (escalation level enc). The split produces 76 features (columns) in X. The resulting split produces 9,792 rows for training and 2,448 rows for testing, meaning 80% of the data is used for training. The stratify = y is applied because the dataset is imbalanced, with the legal escalation class making up only 4% of the data. Without stratification, this class could be underrepresented, or missing entirely, from the test set. Setting random state = 42 ensures the split is reproducible across runs. For scaling, the scaler is fit only on X train, learning the mean and range from the training data, and the same transformation is then applied to X test. Although not all models require scaling, it is applied here as a precaution in case it is needed later in the modelling process.

5. Supervised Model Training, Tuning and Evaluation

Four models were built to predict how serious a customer's complaint would become: no escalation, minor, serious, or legal. The main challenge was that most cases were "no escalation", while legal escalation made up only 4% of the data. To stop models from just ignoring rare cases, all models were set up to treat every category fairly. Three of the four models were also tuned by testing many different settings automatically and keeping the best ones.

- **Logistic Regression** was the starting point. Despite its simplicity, it performed well, with an overall accuracy of 96% on the test set (**Appendix A, Figure 6**).
- **Decision Tree** was the weakest model, scoring 92.6% accuracy. The problem was that it memorised the training data too well, when given new data it hadn't seen before, its performance dropped. It was best at catching legal escalation cases but worse overall. Its main usefulness is that it's easy to explain (**Appendix A, Figures 7 and 8**).
- **Random Forest** scored 94.7% accuracy, sitting in the middle. Performed well at avoiding the memorisation problem. Interestingly, it still couldn't beat the Logistic Regression, which suggests the data itself is relatively straightforward and doesn't need a complicated model. Its feature importance chart was the most useful output; it clearly showed that the risk indicator is the most important predictor, followed by service pressure score and customer engagement score (**Appendix A, Figures 9 and 10**).

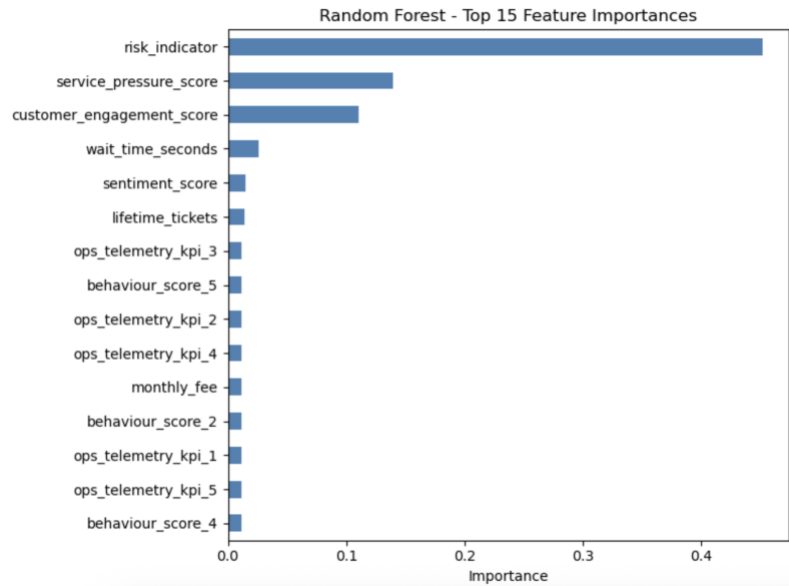


FIGURE 10: RANDOM FOREST - TOP 15 FEATURE IMPORTANCES

- **Gradient Boosting** was the best model, scoring 96.3% accuracy. This made it good at catching legal escalations, improving that score from 0.88 to 0.95, the biggest improvement across all models. The matrix shows exactly where the best model makes mistakes (91 correct, 5 misclassified as serious) (**Appendix A, Figure 11**).

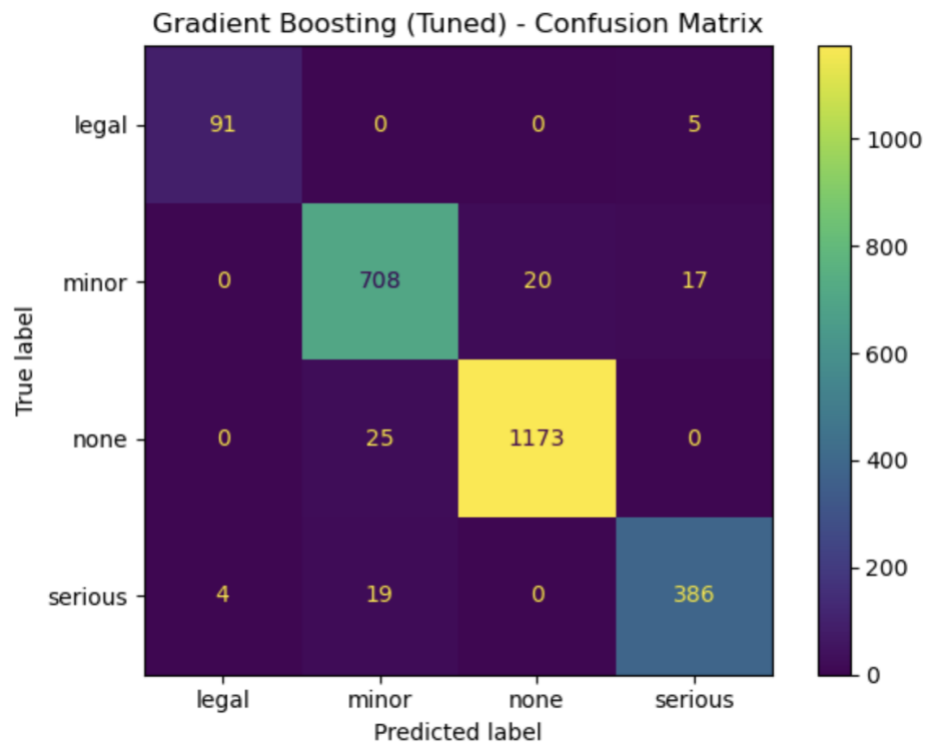


FIGURE 11: GRADIENT BOOSTING CONFUSION MATRIX

The **SHAP analysis** was used to look inside the Gradient Boosting model and understand why it makes certain predictions. The finding was clear and consistent across all four escalation levels: risk indicator and service pressure score are the two features that matter most. High values of both almost always mean escalation is coming. Below those two, wait time, customer engagement, sentiment score, and lifetime tickets also play a role (**Appendix A, Figure 13**).

Overall, every model scored over 92% accuracy, which shows the data contains enough information to predict escalation well. A business could realistically use risk indicator and service pressure score as early warning signals to flag customers before a situation escalates. The model comparison chart showcases each model's performance (**Appendix A, Figure 12**).

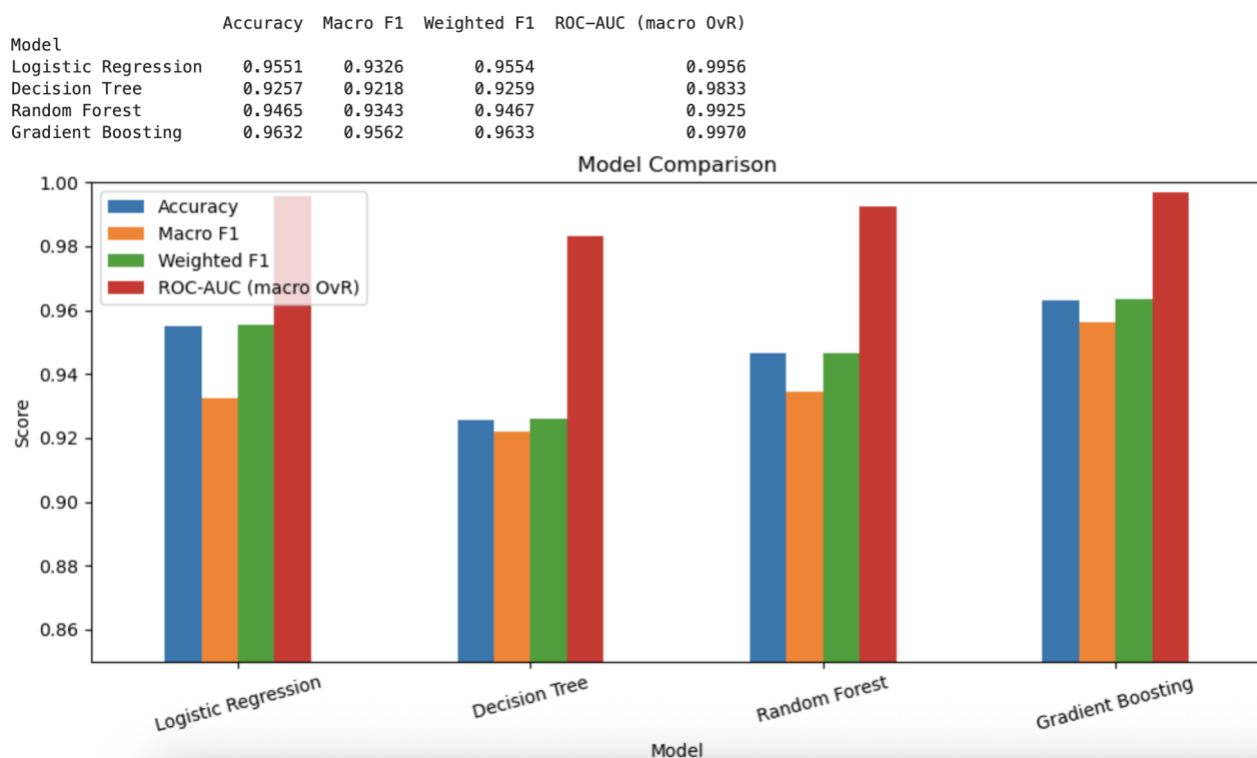


FIGURE 12: MODEL COMPARISON CHART

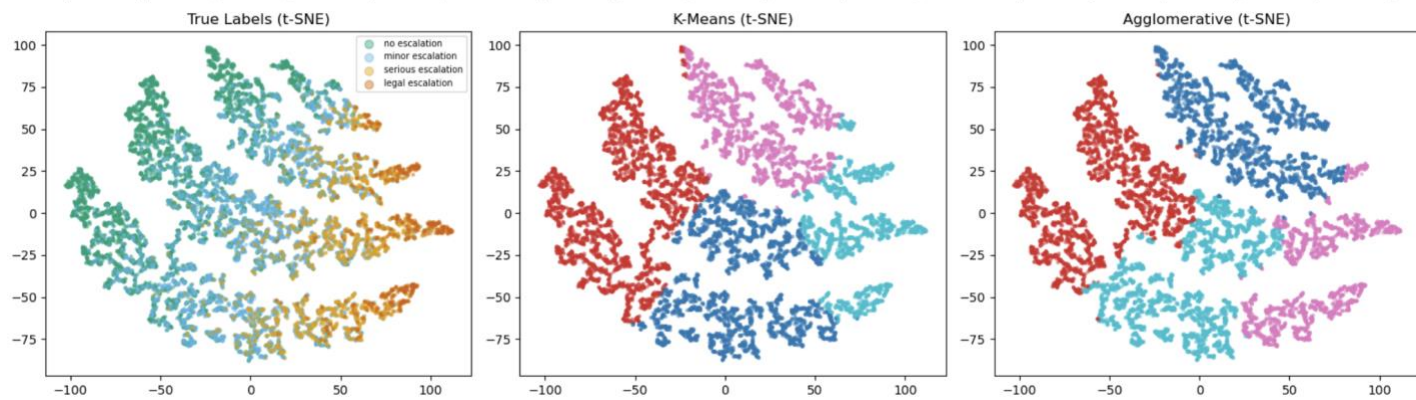
6. Unsupervised Learning Using Clustering Algorithms

The goal is to discover natural customer groupings without using escalation labels and then see whether those clusters align with escalation behavior, using Silhouette to measure separation and ARI to measure label alignment. As shown in [Table 1](#), both K-Means and Agglomerative produce low to moderate scores, while DBSCAN collapses into a single cluster, indicating weak structure and limited alignment with true escalation outcomes. This is expected because escalation risk exists on a spectrum, and the data is broadly distributed rather than clustering into clear, dense groups.

Table 1

Method	Silhouette	ARI vs True Labels
K-Means	0.35	0.21
Agglomerative	0.30	0.18
DBSCAN	N/A	(1 cluster found)

The cluster structure illustrates the comparison between the true escalation labels and the algorithm outputs. Because PCA compresses the data, the resulting scatter plot appears as a cloud, which is expected when groups are not cleanly separable. In contrast, t-SNE better shows relationships and reveals more distinct groupings in the lower row (**Appendix A, Figure 14**).

**FIGURE 14: VISUALISING CLUSTER STRUCTURE (LOWER ROW)**

The cluster to escalation is key because it directly addresses how the discovered clusters relate to escalation levels by showing their composition (**Appendix A, Figure 15**). The heatmap displays the percentage of each escalation level within each K-Means cluster. Notably, cluster 1 contains 84% no escalation cases, clearly identifying a low-risk group, while cluster 3 stands out with 55% serious and 22% legal escalations. Evidence that the model identified a high-risk customer group without seeing the labels. This indicates that the underlying features carry meaningful signals about escalation risk. The K-Means is captured more clearly than the Agglomerative clustering, which produces more blended and less interpretable groups.

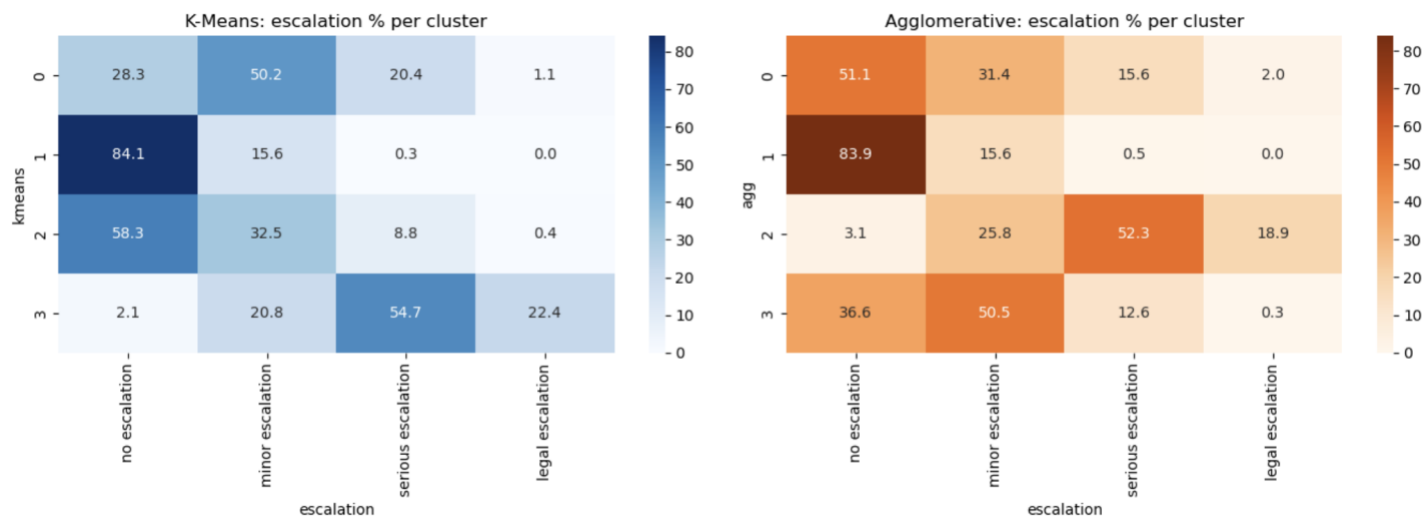


FIGURE 15: VISUALISING CLUSTER TO ESCALATION LEVEL

7. Key Findings

The major key findings from the analysis are:

- Escalation is rare but serious:** Nearly half of all interactions (49%) result in no escalation at all. Only 4% reach legal escalation. This means things are being handled fine day-to-day, but that small percentage of legal cases are costly, damaging, and should be a priority to prevent.
- First contact resolution is the norm:** Data exploration confirmed that 72.2% of interactions were resolved on first contact, with the remaining 27.8% flagged as unresolved, pointing to resolution quality as key for reducing escalation risk.
- The data can predict escalation with 96% accuracy:** The Gradient Boosting model correctly identified the escalation level 96% of the time. This is very high and means the patterns in the data are strong. All the information needed to predict who is about to escalate before it happens is present.
- The top warning signs for escalation:** From the SHAP analysis, the features that matter most shift slightly depending on the outcome being predicted, but the clearest signals across serious and legal escalations are risk indicator, service pressure score, customer engagement score, wait time in seconds, total burden, and sentiment score, with risk indicator and service pressure score consistently sitting at or near the top.
- There is a distinct group of high-risk customers:** The clustering (without being told labels) naturally found a group, K-Means cluster 3, that was 55% serious escalations and 22% legal escalations. This group exists in the data independently. These customers can be identified and targeted prior to escalating further.
- Most customers who escalate have a history:** High lifetime tickets, multiple SLA breaches, and frequent recent interactions are all connected to escalation. Escalating customers seem to have repeated bad experiences building up over time.

In conclusion, the dataset of 12,362 records required substantial cleaning, with misspelled categorical values, duplicate rows, and meaningful missingness across key behavioural fields all addressed before any modelling could take place. The supervised classification results were strong. Gradient Boosting emerged as the best performing model, achieving 96.3% accuracy, a macro F1 of 95.6%, and a ROC-AUC of 99.7%, outperforming Logistic Regression, Decision Tree, and Random Forest across every metric. The fact that even a baseline Logistic Regression model reached 95.5% accuracy suggests linear signal, validating the feature engineering choices made during preprocessing. SHAP analysis identified risk indicator, service pressure score, and customer engagement score as the most consistent influences of serious and legal escalation outcomes. The unsupervised clustering analysis reinforced these findings from a different angle. Without being given any labels, K-Means naturally isolated a cluster in which 55% of customers had experienced serious escalation and a further 22% had reached legal escalation. This confirms that a high-risk customer profile exists structurally within the data and can be detected independently of any model predictions.

Taken together, the findings demonstrate that escalation is not random. It follows identifiable patterns tied to service quality, customer history, and interaction outcomes. The tools developed here provide a foundation for early intervention, allowing attention toward the customers at most risk before situations become severe and costly.